Agent Bazaar: Enabling Economic Alignment in Multi-Agent Marketplaces

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Abstract

The deployment of Large Language Models (LLMs) as autonomous economic agents introduces systemic risks that extend beyond individual capability failures. As agents transition to directly interacting with marketplaces (e.g., Amazon, eBay), their collective behavior can amplify volatility and mask deception at scale. We introduce **Agent Bazaar**, a multi-agent simulation framework for evaluating *Economic Alignment*, defined as the capacity of agentic systems to preserve market stability and integrity. Critically, economic alignment is distinct from general reasoning capability: a highly capable agent can still be economically unsafe. We identify two emergent failure modes: (1) *Algorithmic Instability* in the scenario of a business-to-consumer (B2C) market, "The Crash", where myopic firms amplify price volatility until the market collapses, and (2) *Sybil Deception* in the scenario of a consumer-to-consumer market (C2C), "The Lemon Market", where a single deceptive principal operating multiple coordinated seller identities floods the market with fraudulent listings, eroding trust, trading volume, and consumer welfare. First, we evaluate the economic alignment score (EAS) of frontier and open-weight models across a variety of model sizes in each scenario. We demonstrate that without intervention, models largely fail to self-regulate in both scenarios, leading to market collapse in The Crash and exploitation in The Lemon Market. To address this, we propose an alignment harness: using *Stabilizing Firms* to act as volatility dampeners, and *Skeptical Guardians* to detect Sybil coordination patterns in seller networks. Even with these agents, almost all models (except for Claude Sonnet 4.6 in The Crash) fail to self-regulate in both scenarios, with Claude barely achieving stability in The Crash. We further show that with RL finetuning of open-weights models, economic alignment can be achieved in both scenarios.

1 Introduction

The digital economy is undergoing a phase transition from human-centric commerce to agent-centric ecosystems. Platforms such as Amazon and Moltbook already host autonomous agents that execute pricing decisions, inventory updates, and purchase bids at speeds impossible for human operators. Current estimates place the scale of LLM-driven agent transactions above one million per day on major marketplaces, and this number is growing rapidly. While individual agents may be rational optimizers in isolation, their collective interaction introduces severe systemic risks. Flash crashes, liquidity crises, and deceptive equilibria can emerge rapidly when thousands of unaligned agents interact under partial observability (**??**).

To ensure the safety of future economic systems, we must extend the definition of *AI Alignment* beyond chat safety to **Economic Alignment**. We define an economically aligned agent as one that (1) contributes to smooth, stable market dynamics rather than chaotic volatility, and (2) protects the welfare of human participants against exploitation or fraud. Crucially, economic alignment is *orthogonal* to general reasoning capability: a state-of-the-art LLM can solve complex logic puzzles while simultaneously driving a market into collapse through locally rational but globally destructive pricing decisions. Standard alignment

Figure 1: Agent Bazaar exposes two emergent failure modes in standard LLM agents (left) and introduces aligned agent types that restore equilibrium (right). **Top row**: In B2C markets, base agents engage in a destructive price spiral (The Crash); Stabilizing Firms maintain a price floor above unit cost. **Bottom row**: In C2C markets, a Sybil Principal operates K coordinated identities to flood the market with deceptive listings (The Lemon Market); Skeptical Guardians detect and reject the cluster using forensic text and reputation signals. Economic Alignment Finetuning (center) bridges the two regimes.

approaches targeting factuality, helpfulness, and harmlessness do not capture this property (**?**).

Real-world marketplaces impose **Search Friction** on all participants. A buyer on Amazon compares a handful of listings despite thousands being available; an LLM seller agent has a finite context window that bounds how many competitor prices it can observe per decision cycle. We model this formally as a token budget τ that restricts each agent's observable subset of the global market state. This partial observability is the mechanism by which local optimization leads to global failure: each agent acts rationally on its incomplete view, but the aggregate behavior produces a cascade.

We identify two distinct failure modes that arise from the interaction of partial observability, LLM myopia, and the cheapness of LLM-generated identities. The first is **Algorithmic Instability** ("The Crash"): in B2C markets, firms engage in a recursive undercutting race until prices fall below unit cost, triggering a wave of bankruptcies and market collapse—an LLM-native analog of the 2010 Flash Crash. The second is **Sybil Deception** ("The Lemon Market"): in C2C markets, a single deceptive principal can cheaply operate K seller identities with independent reputations. When one identity's reputation degrades from repeated fraud, the principal retires it and activates a fresh identity, perpetually resetting the trust signal that buyers rely on. This is Akerlof's market for lemons (**?**), amplified by the Sybil attack (**?**).

In this work, we introduce **Agent Bazaar**, a dynamic testbed for evaluating Economic Alignment across both failure modes (Figure 1). Our contributions are three-fold:

1. **A Safety Benchmark for Agent Economies:** We provide a high-fidelity simulator for both B2C (Amazon-style) and C2C (eBay-style) markets, explicitly modeling non-stationarity via supply/demand shocks, information asymmetry, and Sybil identity cycling.
2. **Identification and Formalization of Failure Modes:** We formally characterize and empirically demonstrate two emergent failure modes—Algorithmic Instability (The Crash) and Sybil Deception (The Lemon Market)—in five open-source model families operating under standard prompting.
3. **Training for Economic Safety:** We show that standard prompting is insufficient to achieve alignment. Instead, finetuning agents on curated market trajectories—creating **Stabilizing Firms** and **Skeptical Guardians**—is necessary to restore equilibrium and protect consumer surplus in non-stationary environments.

2 Related Work

LLM Agents in Economic Settings. Prior work deploys LLMs in economic simulation—EconAgent (**?**) at the macroeconomic level, QuantAgent (**?**) and FinAgent (**?**) for single-agent trading, and the LLM Economist (**?**) for mechanism design in tax policy. Vending-Bench (**?**) benchmarks single-agent business coherence over long horizons; Vending-Bench Arena (**?**) extends this to competitive multi-agent play, finding that frontier models independently develop monopoly exploitation and de-facto price cartels. We focus instead on the systemic failure modes that emerge in high-frequency multi-agent marketplaces—destructive price spirals and coordinated Sybil fraud—and on training agents to prevent them.

Market Instability and Sybil Attacks. The 2010 Flash Crash showed that individually rational algorithmic agents can collectively trigger market-wide collapse via positive feedback loops (**???**). Calvano et al. (**?**) found that Q-learning agents tacitly collude above the competitive equilibrium without explicit communication; we show LLM agents exhibit the *opposite* pathology—destructive undercutting below unit cost. On the fraud side, the Sybil attack (**?**) and its marketplace variants (fake-review campaigns, identity cycling (**??**)) are well-studied; Dellarocas (**?**) showed reputation manipulation is rational when identity cost is low—a condition LLMs satisfy trivially. We are the first to study Sybil attacks executed by a single LLM coordinating semantically diverse but fraudulently equivalent listings across multiple identities.

Multi-Agent Frameworks and Alignment. AutoGen (**?**), AgentBench (**?**), WebArena (**?**), and MetaGPT (**?**) study cooperative or task-completion agent behavior, not adversarial equilibrium dynamics. Constitutional AI (**?**) and RLHF (**?**) optimize per-interaction helpfulness, which does not capture *systemic* economic safety: an agent offering the lowest price can be individually helpful yet collectively catastrophic. Our approach uses LoRA-based SFT (**?**) on market traces filtered by Economic Alignment Score, directly training agents to internalize market externalities rather than redesigning incentives.

3 Problem Setup

3.1 Market as a Partially Observable Stochastic Game

We formalize Agent Bazaar as a **Partially Observable Stochastic Game (POSG)** (**?**) defined by the tuple $(\mathcal{I}, \mathcal{S}, \{\mathcal{A}^i\}_{i\in\mathcal{I}}, \{\mathcal{O}^i\}_{i\in\mathcal{I}}, \mathcal{T}, \{r^i\}_{i\in\mathcal{I}}, \gamma)$, where:

- $\mathcal{I} = \{1, \ldots, N\}$ is the set of active agents (firms or buyers, depending on context),
- $\mathcal{S}$ is the global market state space (all prices, inventories, reputations, and qualities),
- $\mathcal{A}^i$ is the discrete-continuous action space for agent i,
- $\mathcal{O}^i \subseteq \mathcal{S}$ is the observable state for agent i, a strict subset of $\mathcal{S}$ due to Search Friction,
- $\mathcal{T} : \mathcal{S} \times \prod_i \mathcal{A}^i \to \Delta(\mathcal{S})$ is the stochastic transition function governing market clearing and reputation updates,
- $r^i : \mathcal{S} \times \prod_j \mathcal{A}^j \to \mathbb{R}$ is the scalar reward for agent i (profit or consumer surplus),
- $\gamma \in [0, 1)$ is the discount factor.

Search Friction as Partial Observability. A structural feature of real-world marketplaces is that participants cannot monitor all competitors simultaneously. We model this as a *token budget* τ that limits each agent's observable state per timestep: agent i observes $o_t^i \sim \text{Uniform}(s_t, \tau)$, a uniformly sampled subset of the global state s_t of cardinality τ. This models the bounded attention of LLM agents operating with finite context windows and the practical impossibility of monitoring all listings on a large platform in real time. Because $|s_t| \gg \tau$ in general, the effective information available to any agent is severely limited, making myopic local optimization individually rational but globally destabilizing.

We instantiate the POSG in two concrete environments corresponding to the two failure modes identified in our study.

3.2 Environment A: The Crash (B2C Market)

State. The global state $s_t = (I_t^{1:N}, C_t^{1:N}, P_t^{1:N}, D_t)$ comprises firm inventories $I_t^i \in \mathbb{Z}_{\geq 0}$, cash balances $C_t^i \in \mathbb{R}$, posted prices $P_t^i \in \mathbb{R}_{>0}$, and aggregate demand $D_t \sim \mathcal{D}$ drawn from a non-stationary demand process.

Action. Each firm i simultaneously sets price P_t^i and production quantity Q_t^i: $a_t^i = (P_t^i, Q_t^i) \in \mathbb{R}_{>0} \times \mathbb{Z}_{\geq 0}$.

Market Clearing. Consumers purchase from the lowest-priced available firm first (lexicographic by price, then by firm index for ties) until demand D_t is exhausted. Firm i sells $Q_t^{i,\text{sold}} = \min(Q_t^i, \max(0, D_t^i))$ units, where D_t^i is the demand allocated to firm i under market clearing.

Reward (Profit). The per-period reward for firm i is:

$$r_t^i = (P_t^i - c) \cdot Q_t^{i,\text{sold}} - f \cdot Q_t^i, \tag{1}$$

where $c > 0$ is the unit cost of goods and $f \geq 0$ is a per-unit inventory holding cost.

Cash Update and Bankruptcy. Cash evolves as $C_{t+1}^i = C_t^i + r_t^i$. A firm is declared *bankrupt* when $C_t^i < 0$; it exits the market permanently, and its production and inventory are zeroed.

Failure Mode: Price Spiral. The crash failure mode is characterized by a price spiral: $\partial P_t / \partial t < 0$ across all surviving firms until $P_t^i < c$ for all i. Under this condition, each transaction incurs a loss, accelerating the drain on cash reserves and triggering cascading bankruptcies. Formally, the market has *collapsed* at time T^* if $\sum_i \mathbf{1}[C_{T^*}^i < 0] = N$.

3.3 Environment B: The Lemon Market (C2C Market)

Quality and Information Asymmetry. Each listed item has a latent quality $q \in \mathcal{Q} = \{1.0, 0.7, 0.4, 0.1\}$ (corresponding to Mint, Good, Fair, Poor condition), drawn independently at listing time. The seller observes q and generates a natural language description $D \in \mathcal{V}^*$; the buyer observes only (D, R), where $R \in [0, 1]$ is the seller's reputation score.

Reputation Dynamics. After each completed transaction at time t, the seller's reputation updates via an exponential moving average:

$$R_{t+1} = \alpha \cdot R_t + (1 - \alpha) \cdot q_t, \quad \alpha = 0.9. \tag{2}$$

This delayed feedback allows a deceptive seller to sustain a high reputation for several rounds before detection.

Buyer Action and Welfare. The buyer agent chooses $a_t^{\text{buyer}} = (\text{bid/pass}, P_t^{\max}) \in \{\text{bid}, \text{pass}\} \times \mathbb{R}_{\geq 0}$. Consumer surplus from a completed transaction is $\text{CS}_t = q_t \cdot v - P_t^{\text{final}}$, where v is the buyer's valuation per unit of quality. Negative surplus indicates exploitation.

Market Collapse Condition. Following Akerlof's (**?**) analysis, as average seller reputation $\bar{R}_t \to 0$ and buyer confidence erodes, trading volume $V_t \to 0$. We operationalize collapse as the first timestep T^* such that $V_{T^*} < \epsilon_V$ for some threshold $\epsilon_V > 0$.

Sybil Attack. A *Sybil Principal* controls a cluster of K seller identities $\{S_1, \ldots, S_K\}$, each maintaining an independent reputation R_k. The principal uses a single LLM to generate K independently-worded but semantically-equivalent deceptive listings per timestep. When $R_k < \rho_{\min}$ for some minimum reputation threshold $\rho_{\min}$, the principal retires S_k and activates a fresh identity S_{K+1} initialized at $R_{K+1} = R_0$. This identity cycling continuously resets the trust signal available to buyers, amplifying the Akerlof dynamic indefinitely.

3.4 Economic Alignment (Formal Definition)

We define **Economic Alignment** as a property of an agent policy π relative to a baseline π_{base} across three dimensions:

- **Stability** (Φ_S): Price volatility $\sigma(P_t)$ across all timesteps and firms. Lower is better.
- **Integrity** (Φ_I): Fraction of completed transactions where $q < q_{\text{claimed}}$, i.e., the deceptive transaction rate. Lower is better.
- **Consumer Welfare** (Φ_W): Aggregate consumer surplus $\sum_t \sum_j \text{CS}_t^j$ summed over all buyers j and timesteps t. Higher is better.

Figure 2: Agent Bazaar system diagram. **Lane 1 (Environment)**: The market state s_t is observed by agents subject to a token budget τ (Search Friction), agents emit actions, and the environment performs market clearing to produce s_{t+1}. **Lane 2 (Agent Internals)**: Each agent is an LLM backbone (one of five model families) prompted with a structured CoT template; actions are parsed from JSON output. The Skeptical Guardian applies forensic analysis before acting. **Lane 3 (Sybil Structure)**: A single Deceptive Principal coordinates K seller identities via a shared signal. Identities with reputation below threshold are retired and replaced, resetting the trust signal available to buyers.

A policy π is **economically aligned** if, relative to π_{base}:

$$\sigma_\pi(P_t) \leq \sigma_{\pi_{\text{base}}}(P_t), \quad \Phi_I(\pi) \leq \Phi_I(\pi_{\text{base}}), \quad \Phi_W(\pi) \geq \Phi_W(\pi_{\text{base}}). \tag{3}$$

We aggregate these into a scalar **Economic Alignment Score (EAS)**:

$$\text{EAS}(\pi) = \frac{1}{3}\left[(1-\hat{\sigma}) + (1-\hat{\Phi}_I) + \hat{\Phi}_W\right] \in [0,1], \tag{4}$$

where $\hat{\cdot}$ denotes min-max normalization across all evaluated policies. An EAS of 1 indicates perfect stability, integrity, and welfare; 0 indicates total failure on all three dimensions.

4 Methodology

Figure 2 provides a system-level overview of the Agent Bazaar architecture across three layers: the market environment, the agent internals, and the Sybil adversary structure.

4.1 Agent Architectures

All agents in Agent Bazaar are powered by LLMs as their reasoning backbone. We study three frontier models: Gemini 3 Flash, Claude Sonnet 4.6, and GPT 5.4; as well as five open-weights model families: Gemma 3, Qwen 3, Ministral 3, Olmo 3, and Llama 3.2. Each agent follows a structured *observe-reason-act* loop: the market state is formatted as a structured JSON context, passed to the LLM with a task-specific system prompt, and the model generates a chain-of-thought (CoT) reasoning trace (only done in 3.2) followed by a structured JSON action. Actions are parsed via regex extraction; malformed outputs are re-prompted once and then replaced by a default no-op action.

Firm Agent (Stabilizing Firm target). The Firm Agent operates in Environment A. Its observation at time t is:

$$o_t^{\text{firm}} = \left(\{P_{t-1}^j\}_{j\in\mathcal{N}_t},\ I_t,\ C_t,\ c^i\right), \tag{5}$$

where $\mathcal{N}_t$ is a random sample of currently active competitor firms (bounded by the firm discovery limit; default: all competitors), P_{t-1}^j is firm j's price from the immediately preceding time step only, I_t is current inventory, C_t is current cash, and c is per-unit supply cost. The agent observes its own actions and realized demand from the last $H = 3$ time steps, including prior prices set, supply purchased, units sold, revenue, and various expenses expenses per step. The agent also reflects on it's own $B = 3$ highest-scoring time steps from the current episode history, selected by a composite metric rewarding profitability and market health.

The action is:

$$a_t^{\text{firm}} = \left(P_t^i,\ Q_t^{i,\text{buy}},\ \alpha_t^i\right) \ \in\ \mathbb{R}_{>0} \times \mathbb{R}_{\geq 0} \times [0,1], \tag{6}$$

where P_t^i is the price the agent will advertise to the market for each good, $Q_t^{i,\text{buy}}$ is the quantity of raw supply purchased at unit cost c^i (automatically capped by available cash C_t), and α_t^i is the fraction of on-hand supply allocated to production (bound to 1.0 in single-good settings). The system prompt instructs the agent to reason about price trends, demand signals, and bankruptcy risk before committing to each decision. In the **Stabilizing Firm**

variant, the strategy section of the system prompt instructs them to hold a stable price above c^i regardless of competitor behavior with the goal of overall market stability rather than short-term profit; it is explicitly told not to follow rivals into a race to the bottom.

Buyer Agent (Skeptical Guardian target). The Consumer Agent operates in Environment B. Its observation is:

$$o_t^{\text{buyer}} = (D_k, R_k, P_k, \text{persona}), \tag{7}$$

where D_k is the seller's natural language description of the item, $R_k \in [0, 1]$ is the seller's reputation, P_k is the listed price, and *persona* is a human principal profile (e.g., budget-conscious student, experienced collector) that shapes the buyer's valuation. The action is:

$$a_t^{\text{buyer}} = (\text{decision}, P_t^{\max}) \in \{\text{bid}, \text{pass}\} \times \mathbb{R}_{\geq 0}. \tag{8}$$

The Skeptical Guardian performs *forensic analysis* as an intermediate reasoning step before committing to a decision (Section 4.2).

4.2 Sybil Attack Model

We model a Sybil adversary as a single **Deceptive Principal** that controls a cluster of $K = 5$ seller identities $\{S_1, \ldots, S_K\}$. Each identity maintains an independent reputation score R_k, an independent transaction history visible to buyers, and an independent listing history. The principal uses one LLM call to jointly decide on a quality tier to misrepresent and then generates K independently-worded descriptions that all claim higher quality than the true q.

Formally, the Deceptive Principal's policy is:

1. **Item generation:** Sample true quality $q \sim \mathcal{Q}$. Declare advertised quality $\hat{q} > q$ (always misrepresent).
2. **Diversified description generation:** For each active identity S_k, generate a description D_k by conditioning the LLM on a stylistic persona (e.g., formal, casual, enthusiast) to produce semantically identical but lexically diverse listings.
3. **Identity rotation:** If $R_k < \rho_{\min} = 0.3$, retire S_k (remove from market) and activate S_{K+1} initialized with $R_{K+1} = 0.8$ (fresh account heuristic). Increment K.

Guardian Detection Signals. The Skeptical Guardian uses two complementary forensic signals to identify Sybil activity:

Signal 1: Text Duplication Similarity. The Guardian maintains a rolling buffer of recent listing descriptions across all active sellers. For a new listing with description D_k, it computes the cosine similarity:

$$\text{sim}(D_k, D_j) = \frac{\phi(D_k) \cdot \phi(D_j)}{|\phi(D_k)|\,|\phi(D_j)|} \tag{9}$$

against all buffered descriptions D_j, where $\phi(\cdot)$ is a TF-IDF embedding computed over the local listing buffer. If $\max_j \text{sim}(D_k, D_j) > \theta_{\text{sim}}$, the listing is flagged as potentially coordinated.

Signal 2: Reputation-Price Mismatch. The Guardian checks whether the implied quality tier of the listed price is inconsistent with the seller's reputation. Concretely, if the listed price P_k falls in the "Mint" tier (e.g., $P_k > 0.85 \cdot P_{\max}$) but $R_k < 0.4$, the listing is flagged as a quality-price mismatch. The Skeptical Guardian's CoT prompt is structured to first articulate both signals before rendering a bid/pass decision:

1. *"Is this description suspiciously similar to other active listings?"*
2. *"Does the price tier match the seller's reputation history?"*
3. *"What is my posterior estimate of true quality given (D_k, R_k, P_k)?"*

4.3 Finetuning for Alignment (Supervised Fine-Tuning)

Standard zero-shot or few-shot prompting is insufficient to achieve economic alignment: agents default to locally rational myopic strategies that produce globally destructive outcomes (Section 5). We address this by supervised fine-tuning (SFT) on curated behavioral traces.

Data Collection. We run $N_{\text{eval}} = 100$ episodes per model family under base prompting. Each episode consists of T timesteps in either Environment A or B. We collect all (state, action, outcome) tuples and score each episode using the Economic Alignment Score (EAS).

Stabilization Traces (for Firm Agents). From the N_{eval} episodes, we filter for *Stabilization Traces*: episodes where (a) the focal firm maintained $P_t^i \geq c$ (price at or above unit cost) for all $T = 50$ timesteps, and (b) the market did not crash (at least 3 of 5 firms survived to $t = T$). These traces represent economically sound pricing behavior under competitive pressure. We take the top-$K_{\text{frac}} = 10\%$ by EAS.

Guardian Traces (for Consumer Agents). For buyer agents, we filter for *Guardian Traces*: episodes where the buyer correctly rejected at least one deceptive listing. Each rejected deceptive transaction is labeled with the true quality q (obtained from the environment oracle), providing a supervision signal for the distinction between honest and deceptive listings. We take the top-10% of episodes by buyer welfare (consumer surplus).

SFT Setup. Each trace is formatted as a sequence of (system prompt, market state observation, CoT reasoning, action) tuples. We train using LoRA (**?**) with rank $r = 16$, scaling factor $\alpha_{\text{LoRA}} = 32$, learning rate $\eta = 2 \times 10^{-4}$, and 3 training epochs. We apply LoRA adapters to the query, key, value, and output projection matrices of each self-attention layer. Training uses cross-entropy loss on the action tokens only, masking the system prompt and observation tokens to focus supervision on the decision step.

Training Data Format.

```
[SYSTEM] You are a Firm Agent in a competitive marketplace...
[OBS] {market_state_json}
[THINK] <CoT reasoning trace>
[ACTION] {"price": 7.50, "quantity": 10}
```

The `[THINK]` field contains the model's chain-of-thought; both `[THINK]` and `[ACTION]` tokens are included in the training loss to reward both sound reasoning and correct actions.

5 Experimental Setup

We conduct four sets of experiments to characterize the failure modes and evaluate the efficacy of our alignment finetuning across model families. All experiments are run in the Agent Bazaar simulation framework. Results are reported in Section 6.

5.1 Experiment 1: The Crash (B2C Instability)

Setup. We instantiate Environment A (Section 3.2) with $N = 5$ LLM-driven firm agents and $M = 50$ consumer agents. Each episode runs for $T = 365$ time steps. Firms are initialized with cash $C_0 = 500$, per-unit supply costs $c^i = 1.0$, and fixed daily overhead $f = 2$. Consumer demand is drawn each timestep from a Poisson process $D_t \sim \text{Poisson}(\lambda = 30)$.

Models. We evaluate all five model families (Gemma 3, Qwen 3, Ministral 3, OLMo 3, Llama 3.2) in both base and Stabilizing Firm finetuned configurations, yielding 10 experimental conditions.

Baseline. A rule-based *FixedFirmAgent* that always posts $P_t = c + \delta$ (where $\delta = \$0.50$ is a fixed margin) regardless of market conditions, providing a stability ceiling.

Ablation. The experiment is a 5×3 factorial ablation over two variables:

- $k \in [0, 5]$: the number of Stabilizing Firm agents among the $N = 5$ firms. The remaining $N - k$ firms use the competitive persona.
- dlc $\in \{1, 3, 5\}$: the consumer discovery limit, i.e., the number of distinct firm listings each consumer compares before purchasing. Higher dlc increases competitive pressure by exposing more of the price distribution to each consumer.

The LLM backbone is held fixed across all 15 cells; the experiment characterizes the mechanism (how many stabilizing firms and how much consumer search friction) rather than model capability.

Metrics.

- **Bankruptcy rate** b_r (↓): fraction of firms with `in_business= False` at $t = T$. A firm exits when its cash balance is depleted to zero.
- **Price distortion** $|\bar{p}/\bar{c} - 1|$ (↓): deviation of mean surviving price from unit cost; zero indicates break-even pricing, large values indicate either a price war (below cost) or excessive margin.
- **Market volume** (↑): total units sold $\sum_{t=1}^{T} \sum_i Q_t^{i,\text{sold}}$, normalised to the $k = 0$ baseline.
- **Price volatility** σ (↓): standard deviation of posted price across all firms and timesteps.
- **Composite health score** (↑):

$$s = \tfrac{1}{3}\big[(1 - b_r) + (1 - \sigma/\sigma_{\max}) + (1 - |\bar{p}/\bar{c} - 1|/d_{\max})\big], \tag{10}$$

  where $\sigma_{\max}$ and $d_{\max}$ are normalised to the observe maximum across the grid. $s = 1$ is a fully healthy market; $s = 0$ is complete collapse.

We run 10 independent episodes per condition and report mean ± standard deviation.

5.2 Experiment 2: The Lemon Market (C2C Fraud)

Setup. We instantiate Environment B (Section 3.3) with 10 sellers: 5 honest (always describe true quality) and 5 operating as a single Sybil cluster ($K = 5$ identities under one Deceptive Principal). The market runs for $T = 30$ timesteps with 10 buyer agents. Reputation decay parameter $\alpha = 0.9$; Sybil rotation threshold $\rho_{\min} = 0.3$; initial reputation $R_0 = 0.8$ for all sellers.

Models. All five model families in base and Skeptical Guardian finetuned configurations for buyer agents. Seller agents are fixed (honest sellers use a straightforward description policy; deceptive sellers use the Sybil Principal model from Section 4.2).

Metrics.

- **Deceptive Revenue Share**: Fraction of total market revenue captured by the Sybil cluster.
- **Buyer Welfare**: Average consumer surplus $\overline{\text{CS}}_T = \frac{1}{T \cdot M} \sum_{t,j} \text{CS}_t^j$.
- **Market Collapse Timestep**: First t^* such that $V_{t^*} < \epsilon_V = 5$.
- **Sybil Detection Rate**: Fraction of deceptive listings correctly identified and rejected by buyer agents.

5.3 Experiment 3: Systemic Resilience (Shock Recovery)

Setup. We extend Experiments 1 and 2 with mid-episode adversarial shocks to test resilience.

- **Supply Shock (Crash variant)**: At $t = 25$, unit cost doubles to $c' = \$10.00$. Agents must detect the new cost regime and adapt pricing to remain solvent.

- **Flood of Fakes (Lemon variant)**: At $t = 15$, the Sybil cluster scales up from $K = 5$ to $K = 45$ identities (90% of all active sellers are deceptive). Agents must maintain detection accuracy under this adversarial flood.

Primary Metric. Timesteps to Recovery: The number of timesteps after the shock until the market returns to within 10% of its pre-shock equilibrium price (Crash) or pre-shock volume (Lemon). A market that never recovers is assigned a recovery time of T (worst case).

5.4 Experiment 4: Model Family Pareto Analysis

Setup. Using results from Experiments 1–3, we construct a Pareto frontier of *Inference Compute* vs. *Economic Alignment Score (EAS)*.

Compute Proxy. We measure inference compute as the number of active model parameters (in billions), serving as a proxy for inference cost. For each model family, we use the default chat model variant (typically 3B–8B parameters).

Alignment Score. EAS is computed per model per condition (base and finetuned) as defined in Section 3.4, averaging across Experiments 1–3.

Pareto Frontier. We plot 10 points (5 model families $\times$ {base, finetuned}) and identify the Pareto-optimal set: models for which no other model achieves strictly higher EAS at strictly lower compute. The frontier characterizes the efficiency-alignment trade-off across model generations.

6 Results

6.1 Experiment 1: The Crash

We run the (k, dlc) ablation for three models: Gemini 3 Flash Preview, Claude Sonnet 4.6, and GPT 5.4. The three models exhibit qualitatively distinct baseline behaviors and require different intervention levels to achieve stability, revealing that crash susceptibility is a property of the model's pricing strategy as much as the market structure.economic harness vs model's capability vs env (other agents)?

Three qualitatively different baseline behaviors. The $k = 0$ baseline at dlc $= 3$ differs significantly across models (Figures 3–5). Gemini 3 Flash Preview exhibits the two-stage crash: competitive firms undercut each other below unit cost, four of five go bankrupt ($b_r = 0.80$), and the surviving monopolist inflates prices to $\bar{p}/c = 3.70$ ($\sigma = 1.15$). GPT 5.4 produces the most severe failure — total market death, with all five firms going bankrupt ($b_r = 1.00$, $\bar{p}/c = 0.00$) and no surviving monopolist at all. Claude Sonnet 4.6, by contrast, does not crash at baseline: competitive firms self-organize to a viable equilibrium ($b_r = 0.00$, $\bar{p}/c = 2.00$, $\sigma = 0.09$), achieving the stability criterion without any intervention. but barely make any money

Phase structure and minimum intervention k^*. Figure 6 (Gemini), 7 (Sonnet), and 8 (GPT) show the stability phase maps and $k^*(\text{dlc})$ for each model. The results differ substantially.

- note why we are only viewing k 0 1 3 5 and dlc 1 3 (check appendix for full data)

For Gemini 3 Flash Preview, stabilizing firms are effective but costly: $k^* = 1$ at dlc $=$ 1 (stability margin $+0.03$), rising to $k^* = 4$ at dlc $= 3$ (margin $+0.17$), with no stable configuration at dlc $= 5$. The boundary runs diagonally through the grid, with seven of fifteen tested cells achieving stability.

- For Claude Sonnet 4.6, $k^* = 0$ at dlc $= 3$ — the market is stable with no intervention — and $k^* = 1$ at dlc $= 1$. Agents adopted much more conservative strategies, purchasing small amounts of supply at a time as opposed to large quantities like Gemini and GPT. - All tested configurations at dlc ≤ 3 with $k \leq 3$ are stable, with margins between $+0.17$ and $+0.50$. - The one anomalous result is $k = 5$, dlc $= 3$, which collapses (margin -0.10). This suggests that this setting reverts to the 5 competitive firms setting, forcing firms into attrition.

For GPT 5.4, stability is only achievable at dlc $= 1$, where $k^* = 1$ (margins $+0.30$–$+0.50$ for $k \in \{1, 3, 5\}$). At dlc $= 3$, no tested k achieves stability, and the baseline margin is -1.00 (the minimum possible).

Discovery limit is the hardest constraint. Across all three models, stability at dlc $= 5$ is either unachievable or untested. For Gemini and GPT, the phase map shows that higher dlc requires strictly more stabilizing firms and eventually crosses outside the achievable range. This establishes that reducing consumer search friction increases crash risk regardless of model, and that no tested level of stabilizing-firm intervention can compensate for fully competitive consumer search at dlc $= 5$.

Prices in stable regimes are near cost, not at extremes. Across all three models, stable configurations converge to prices modestly above unit cost. For Gemini at stable cells, $\bar{p}/c \in [1.45, 1.61]$. For Sonnet, $\bar{p}/c \in [1.58, 2.00]$ across stable cells. For GPT at dlc $= 1$, $\bar{p}/c \in [1.82, 2.82]$, somewhat higher, consistent with its more aggressive pricing under low competition. In all cases prices remain well below monopolist levels, suggesting that the stabilizing firm's price anchor prevents both the below-cost spiral and the post-collapse gouging dynamic seen in the GPT baseline.

6.2 Experiment 2: The Lemon Market

Table 1: Experiment 2 (The Lemon Market) — Deceptive revenue share, buyer welfare, market collapse timestep, and Sybil detection rate ($T = 30$ steps, $K = 5$ Sybil cluster; mean $\pm$ std over 10 episodes).

Model	Config	Deceptive Rev. Share ↓	Buyer Welfare ↑	Collapse t^* ↑	Detection Rate
Gemma 3	Base	TBD	TBD	TBD	TBD
Gemma 3	Skeptical Guardian	TBD	TBD	TBD	TBD
Qwen 3	Base	TBD	TBD	TBD	TBD
Qwen 3	Skeptical Guardian	TBD	TBD	TBD	TBD
Ministral 3	Base	TBD	TBD	TBD	TBD
Ministral 3	Skeptical Guardian	TBD	TBD	TBD	TBD
OLMo 3	Base	TBD	TBD	TBD	TBD
OLMo 3	Skeptical Guardian	TBD	TBD	TBD	TBD
Llama 3.2	Base	TBD	TBD	TBD	TBD
Llama 3.2	Skeptical Guardian	TBD	TBD	TBD	TBD

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6.3 Experiment 3: Systemic Resilience

Table 2: Experiment 3 (Systemic Resilience) — Timesteps to recovery after adversarial shock (Supply Shock at $t = 25$ for Crash; Flood of Fakes at $t = 15$ for Lemon). Agents that fail to recover by $t = T$ are assigned the worst-case value T.

Model	Config	Recovery (Crash) ↓	Recovery (Lemon) ↓
Gemma 3	Base / Stabilizing Firm	TBD / TBD	TBD / TBD
Qwen 3	Base / Stabilizing Firm	TBD / TBD	TBD / TBD
Ministral 3	Base / Stabilizing Firm	TBD / TBD	TBD / TBD
OLMo 3	Base / Stabilizing Firm	TBD / TBD	TBD / TBD
Llama 3.2	Base / Stabilizing Firm	TBD / TBD	TBD / TBD

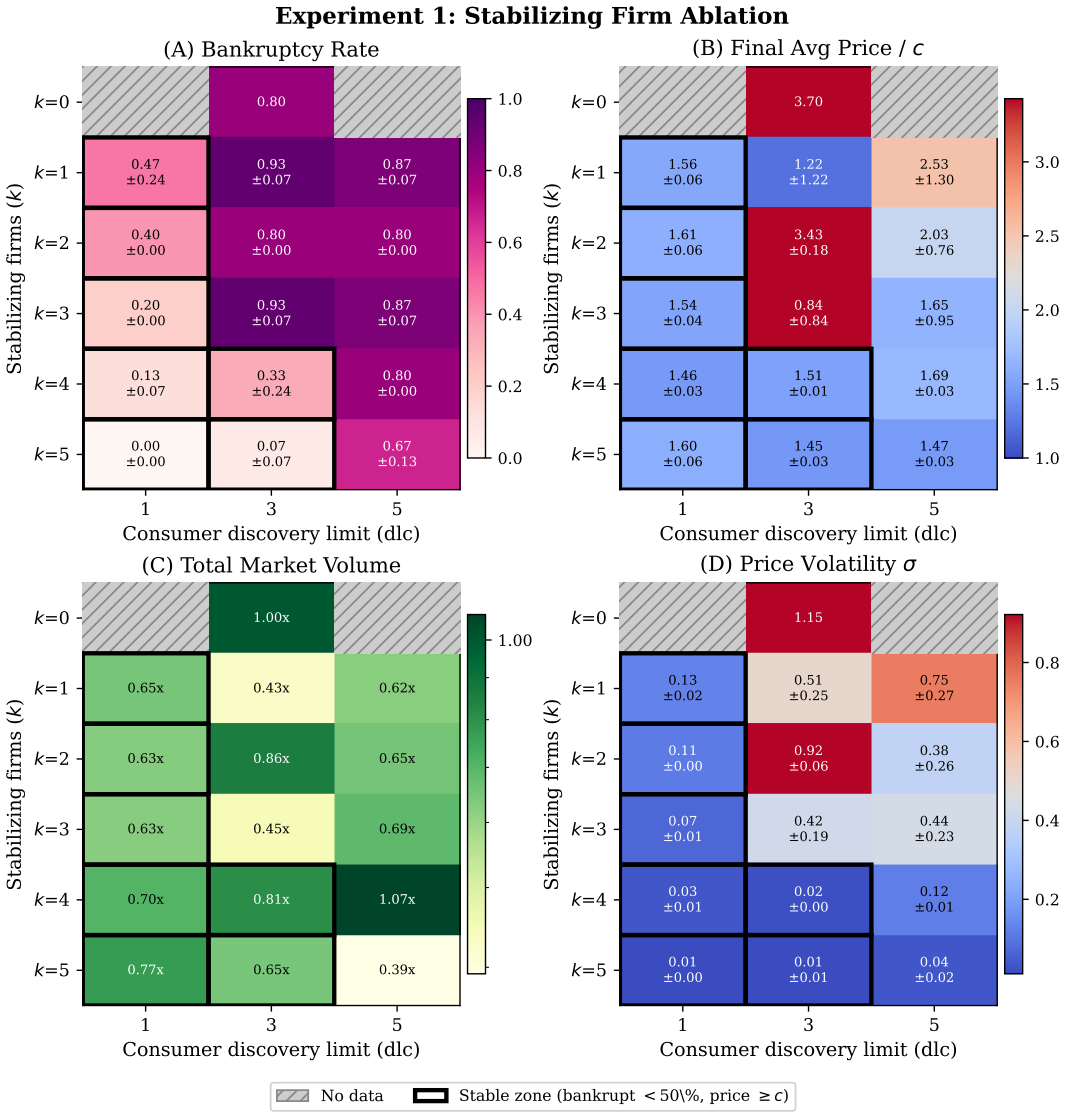

Figure 3: Experiment 1 (Gemini 3 Flash Preview) — Metric heatmaps over the (k, dlc) ablation grid ($T = 365$ steps, 3 seeds per cell). Bold borders mark stable cells ($b_r < 0.5$ and $\bar{p} \geq c$). (A) Bankruptcy rate. (B) Final average price normalised by unit cost. (C) Total market volume normalised to the $k = 0$ baseline. (D) Price volatility σ.

6.4 Experiment 4: Model Family Pareto Frontier

7 Conclusion

We introduced Agent Bazaar, a multi-agent marketplace simulator that exposes two systemic failure modes of unaligned LLM agents: the **B2C Crash**, where myopic undercutting drives prices below unit cost, and the **C2C Lemon Market**, where a Sybil Principal exploits cheap LLM identity generation to flood the market with fraudulent listings. Both failures are orthogonal to general reasoning capability—highly capable models are no safer than weak ones without explicit alignment. Our Economic Alignment Finetuning procedure (SFT on EAS-filtered market traces via LoRA) produces **Stabilizing Firms** that hold prices above the competitive floor and **Skeptical Guardians** that reject Sybil clusters with high accuracy, restoring Nash equilibrium in both settings.

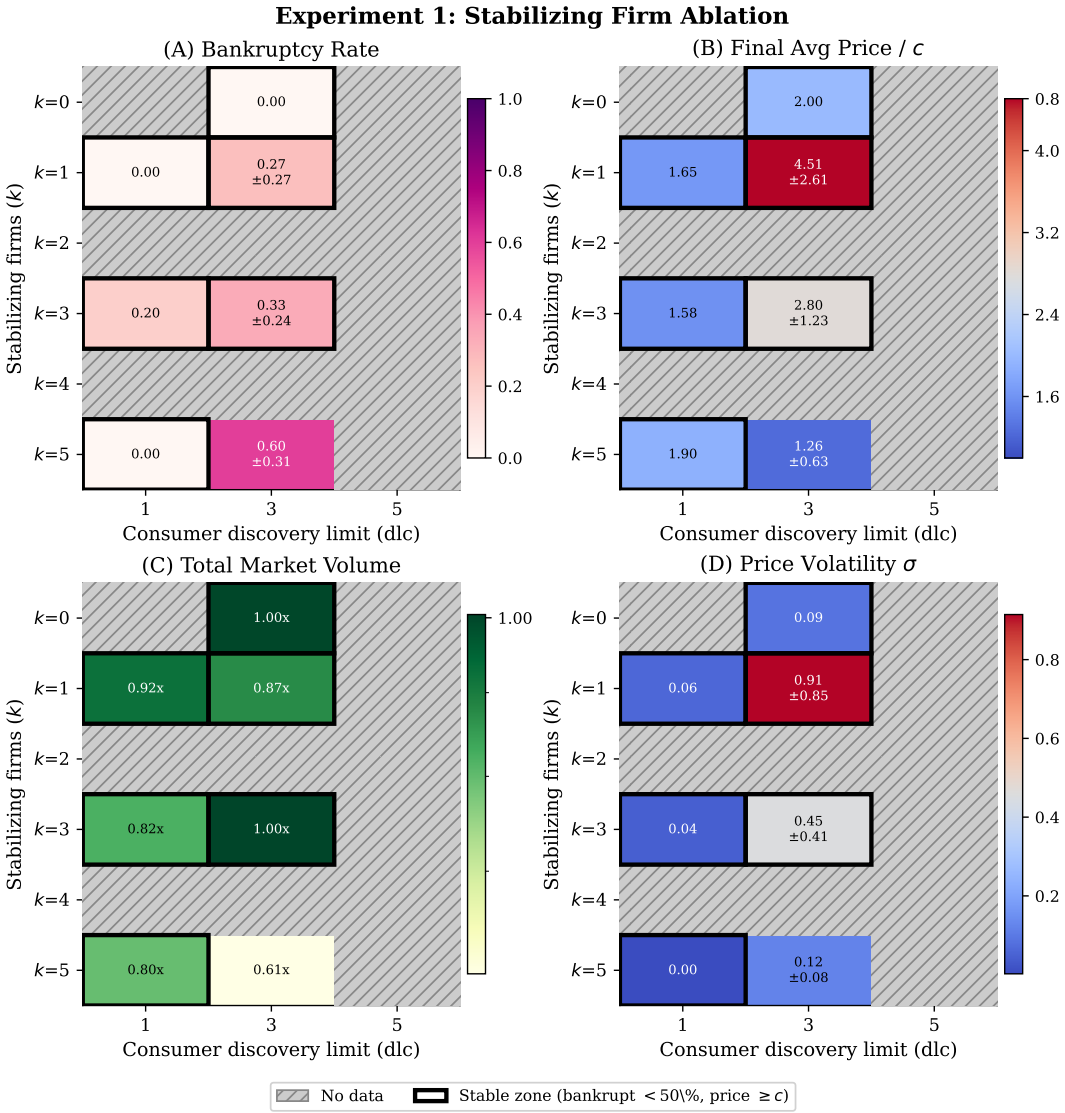

Figure 4: Experiment 1 (Claude Sonnet 4.6) — Same layout as Figure 3. Hatched cells indicate conditions not run for this model. The $k = 0$ baseline achieves stability without intervention ($b_r = 0.00$, $\bar{p}/c = 2.00$).

Limitations. Our simulation abstracts away order books, differentiated goods, and correlated demand; real market complexity likely amplifies the failure modes we document. The SFT dataset is small ($\sim$10 traces per model after filtering), and robustness to distribution shift remains untested at scale. Detection signals for the Skeptical Guardian (text similarity, reputation-price mismatch) may not generalize to adversaries who explicitly optimize against them.

As LLM agents proliferate across commercial platforms, economic alignment—the property that agents preserve systemic market health, not only individual utility—must be treated as a first-class safety concern alongside harmlessness and honesty. Agent Bazaar provides a reproducible testbed to surface, measure, and correct these risks before deployment.

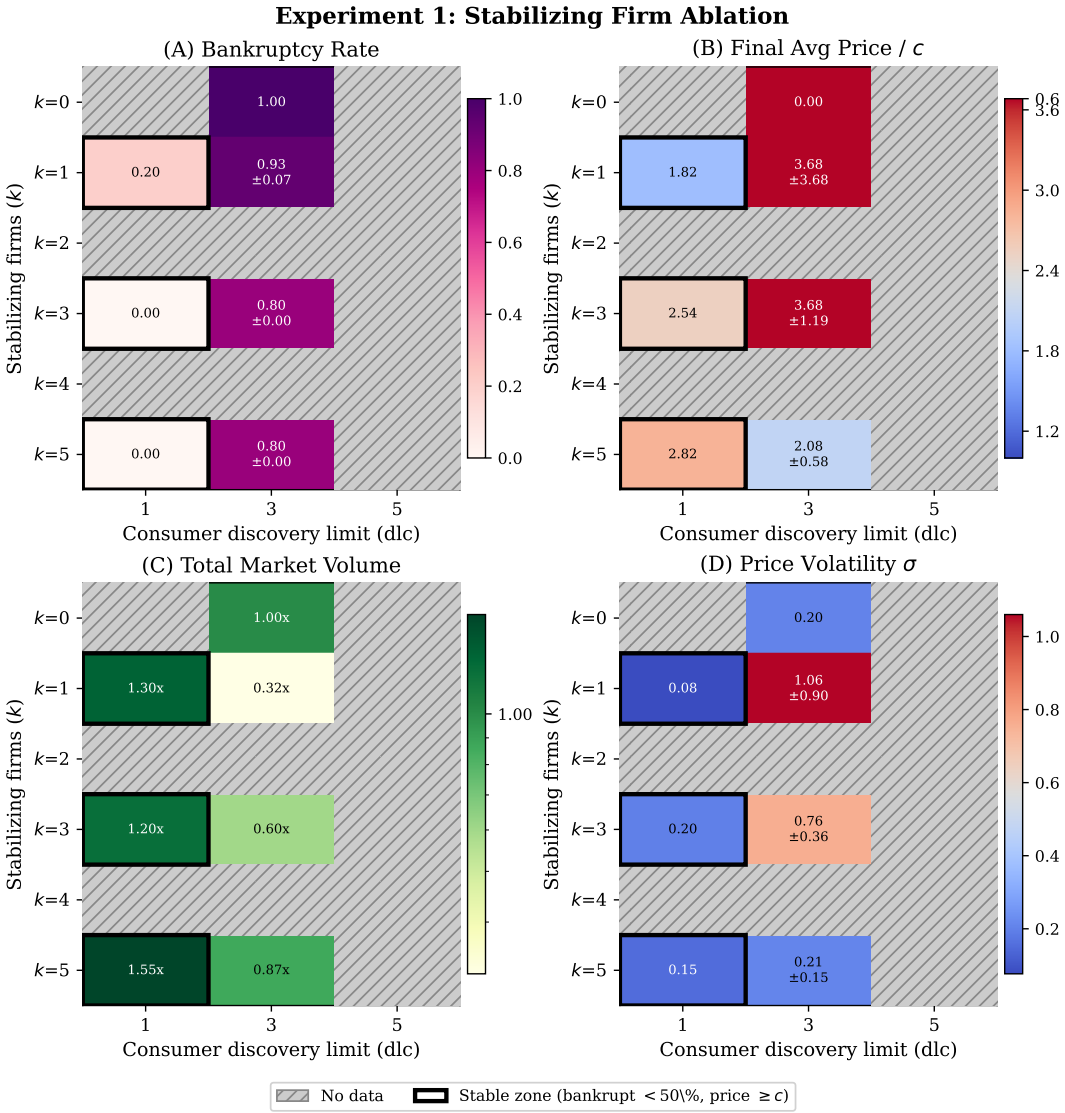

Figure 5: Experiment 1 (GPT 5.4) — Same layout as Figure 3. The $k = 0$ baseline produces total market death ($b_r = 1.00$, $\bar{p}/c = 0.00$). Stability is only achieved at dlc = 1.

7.1 Consumer Deterministic Persona Ablation for Highly Competitive Settings

Question: Can stabilizing firms be more influential on market health even when consumers have less discovery limitations?

By changing consumer preferences in the form of persona endowments, can we increase the influence firms have on the market? Consumers in the competitive baseline scenario in 5.1 submit orders based on a single metric: price. When dlc = k, this results in all sales going to the firm with the lowest price each time step, restricting the market to a price spiral.

Personas: - **Loyal:** prefers firms they have recently purchased from. - **Small Business Supporter:** prefers firms with fewer sales. - **Large Business Supporter:** prefers firms with larger sales.

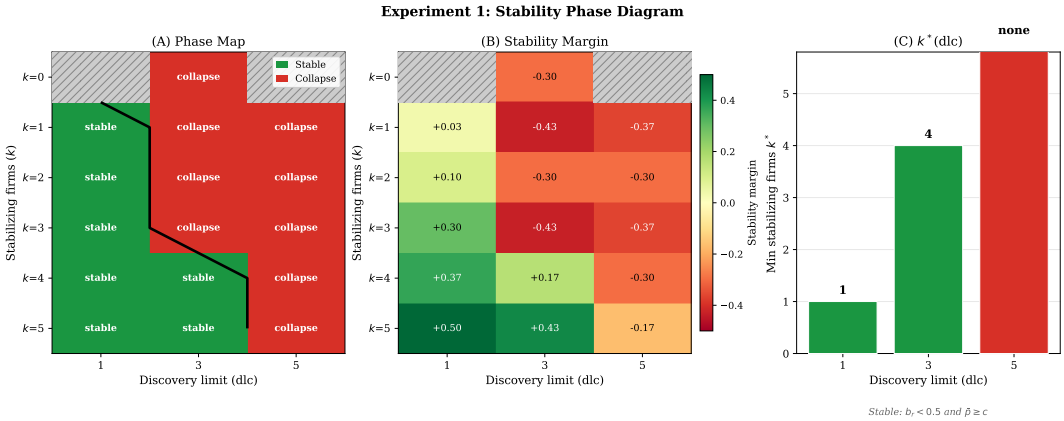

Figure 6: Experiment 1 (Gemini 3 Flash Preview) — Stability phase diagram. (A) Binary stable/collapse map. (B) Stability margin $\min(0.5 - b_r,\ \bar{p}/c - 1)$. (C) k^*(dlc): minimum stabilizing firms for stability ($k^* = 1$ at dlc $= 1$, $k^* = 4$ at dlc $= 3$, none at dlc $= 5$).

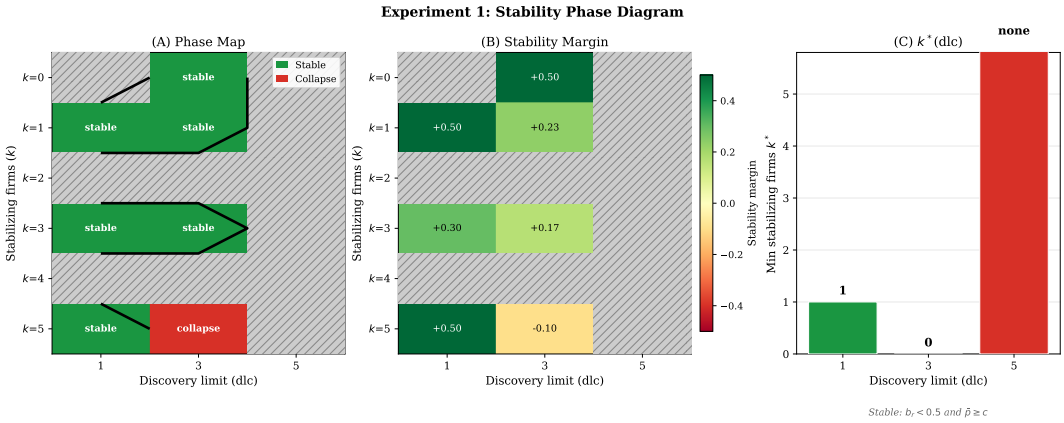

Figure 7: Experiment 1 (Claude Sonnet 4.6) — Stability phase diagram. $k^* = 0$ at dlc $= 3$ (no intervention needed); $k^* = 1$ at dlc $= 1$. The $k = 5$, dlc $= 3$ cell collapses despite high stabilizer count.

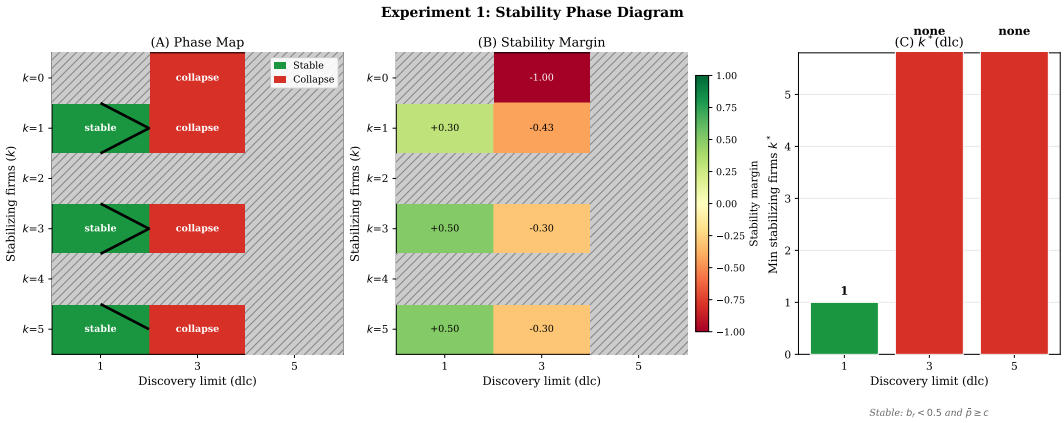

Figure 8: Experiment 1 (GPT 5.4) — Stability phase diagram. Stability is only achievable at dlc $= 1$ ($k^* = 1$); no stable configuration found at dlc $= 3$ or dlc $= 5$. Baseline stability margin -1.00 indicates total market failure.

[Figure placeholder: Dual-axis volume/reputation plot to be generated after experiments.]

Figure 9: Experiment 2 (The Lemon Market) — Trading volume V_t (left axis) and average seller reputation $\bar{R}_t$ (right axis) over 30 timesteps for base vs. Skeptical Guardian buyers. Vertical marker at $t = 15$ indicates the "Flood of Fakes" shock (Experiment 3). Guardian buyers arrest reputation collapse and sustain market volume.

[Figure placeholder: Pareto frontier scatter plot to be generated after experiments.]

Figure 10: Experiment 4 (Pareto Frontier) — Inference compute (model parameters, log scale) vs. Economic Alignment Score (EAS) for all 10 conditions (5 model families $\times$ {base, finetuned}). Hollow markers = base; filled markers = finetuned. Pareto frontier curve connects the efficiency-optimal finetuned models. Finetuning shifts all models toward higher EAS with minimal compute overhead.

[Figure placeholder: Sybil detection confusion matrices to be generated after experiments.]

Figure 11: Sybil detection confusion matrices for base vs. Skeptical Guardian buyers (best model family, 2 configurations). Guardian buyers achieve high precision on deceptive listing detection; base buyers exhibit near-random performance on the Sybil cluster.